

Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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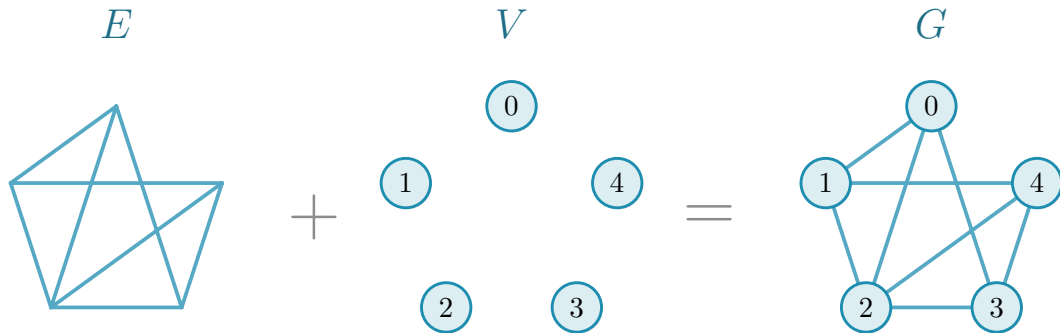
Maximising edge density in graphs
under connectivity constraints



translation

networks with **many links** and **low connectivity**

Graph

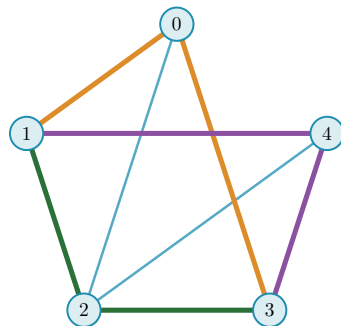


Erdős Problem 915

$$\ell_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \lambda_G^{\max} \leq m-1 \right\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v)$$

$\lambda_G(u, v)$ = the largest number of u - v routes that share **no edge**: one colour each

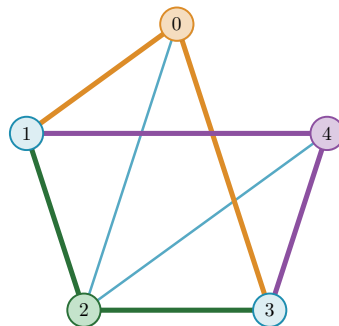


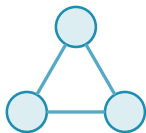
The same question, vertex by vertex

$$k_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \kappa_G^{\max} \leq m-1 \right\}$$

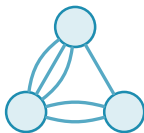
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v)$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**

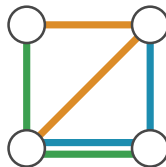




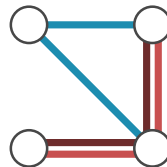
simple graph



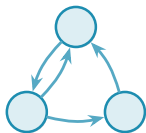
multigraph



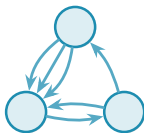
hypergraph



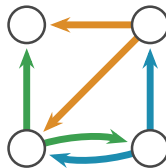
multihypergraph



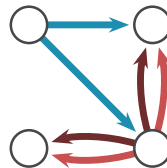
simple digraph



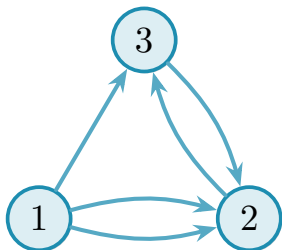
directed multigraph



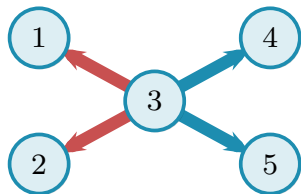
directed hypergraph



directed multihypergraph



$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$



list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

$$\begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

INPUT

||

directed multigraph

max flow
over every pair

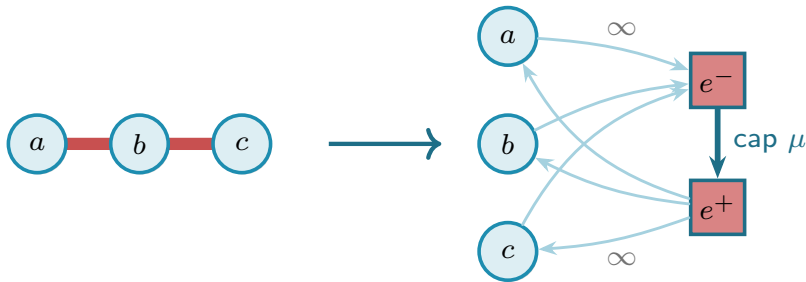
$$\lambda_G^{\max} = 3$$

OUTPUT

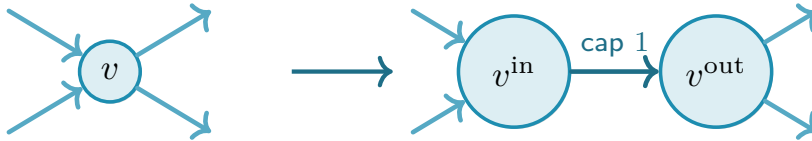
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph



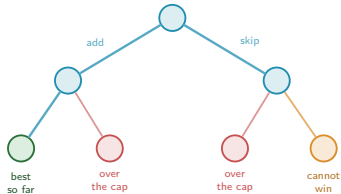
$$\kappa \rightarrow \lambda$$



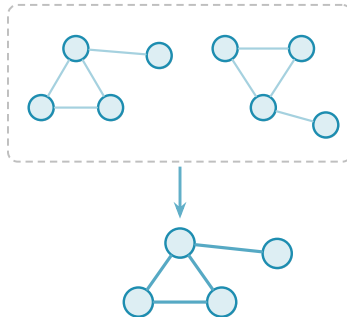
every intermediate vertex is split this way
the flow itself runs from s^{out} to t^{in}

Efficiency: three ways to make the sweep finish

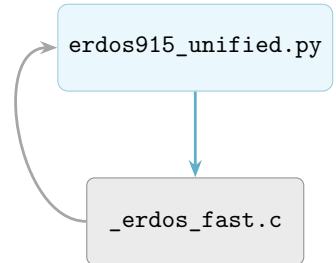
pruning



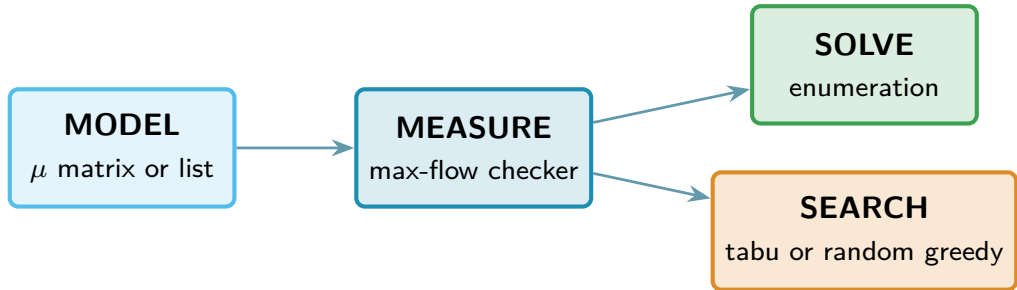
generating search



C: faster max flow

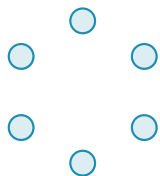


Solving versus searching

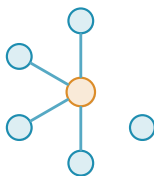


The search: one number per graph

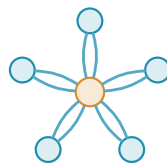
$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$



$$\Phi = 0$$



$$\Phi = -4$$



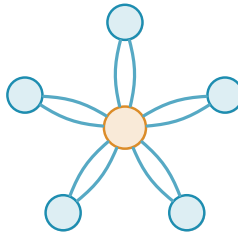
$$\Phi = -10 \text{ at } m = 3$$

Extremal families the search rediscovers



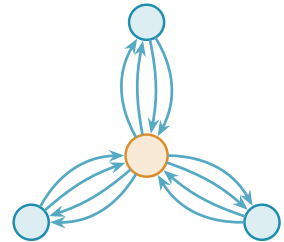
tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

$$n = 6, m = 3, L_3(n) = 10$$

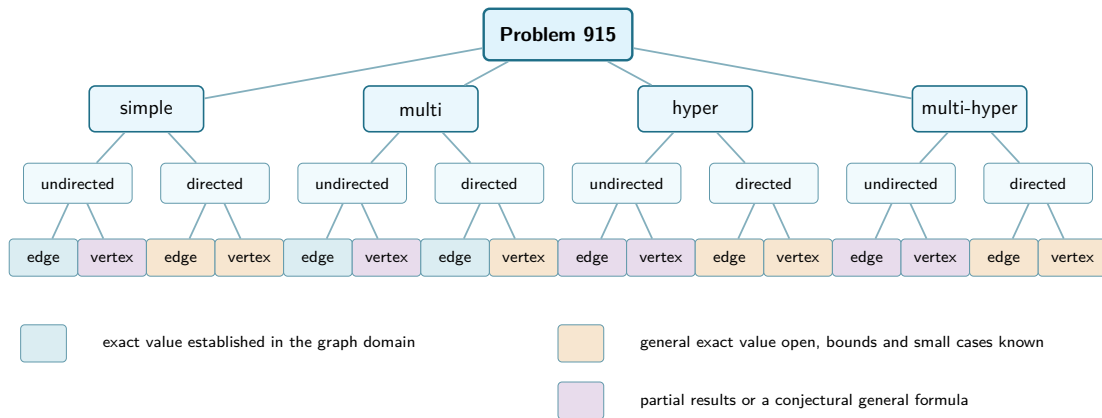


double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

examples of the families, not the saved search witnesses

Status of the sixteen variants



Thank you

`github.com/louisevandenbruwaene/thesis`

Backup: the sixteen variants

WHAT IS KNOWN ABOUT A BOUND

PROVED

established, with a proof the author has checked

PROPOSED

conjecture with a proposed proof the author has not checked line by line

OPEN

no general value established

WHAT A CONSTRUCTION IS WORTH

EXTREMAL

attains the maximum in the stated range

NEAR-EXTREMAL

the densest family reached here, with no matching upper bound

ASYMPTOTICALLY EXTREMAL

matches the leading term of the upper bound

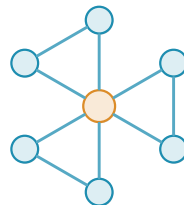
Simple graph · edge

$$\ell_m(n)$$

UPPER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED** Mader, via Gomory–Hu

LOWER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED** for every $n \geq m$

- the two meet for $n \geq m$, so the value is exact there
- for $n < m$ it is $\binom{n}{2}$: not even K_n carries m routes



hub-and-spokes

EXTREMAL

COMPUTED Tabu search returns the tree at $n = 6$, $m = 2$: $5 = \ell_2(6)$.

Simple graph · vertex

$$k_m(n)$$

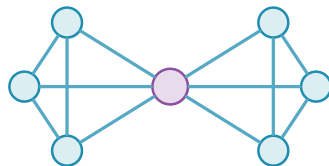
UPPER $k_m(n) = \ell_m(n)$ for $m \leq 4$, $k_5(n) = \lfloor \frac{8n}{3} \rfloor - 4$ for $n \geq 14$ **PROVED**

$m \geq 6$: **OPEN**

LOWER $c_m > m/2$ for $m \geq 5$ **PROPOSED** under the block formulation

- at $m = 5$ the two agree to $n = 13$, first gap 33 against 32 at $n = 14$
- $c_m = \lim_n k_m(n)/(n-1)$, and $m/2 < c_m \leq 2(m-1)$ for $m \geq 5$

COMPUTED Block sweeps reach small blocks only, and the recursive constructions in the literature show why they cannot fix c_m .



2-connected blocks at a cut vertex **NEAR-EXTREMAL**

Simple digraph · arc

$$\ell_m^{\text{dir}}(n)$$

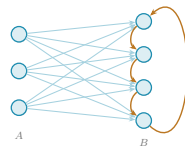
UPPER $M(n) = \max(2(n-1), \lfloor \frac{n^2}{4} \rfloor)$ at $m = 2$ **PROVED**

$\lfloor \frac{n^2}{4} \rfloor + 4(m-1)(n-1)$ at every n **PROVED**

LOWER $\frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1)$ **PROVED**

- for fixed $m \geq 3$, $\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$; the linear coefficient is open
- this family refutes the first guess $\max(m(n-1), \lfloor \frac{n^2}{4} \rfloor)$ at $m = 3, n = 10$

COMPUTED Enumeration confirms $m = 2$ through $n = 7$. Tabu returns 6, 10 and 12 at $n = 4, 6$ and 7.



augmented bipartite digraph

ASYMPTOTICALLY EXTREMAL

Simple digraph · vertex

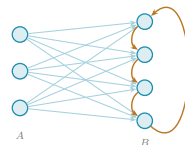
$$k_m^{\text{dir}}(n)$$

UPPER $M(n)$ at $m = 2$ **PROVED**

$$\frac{n^2}{4} + \Theta_m(n) \text{ for fixed } m \geq 3 \quad \textbf{PROVED}$$

LOWER the same bipartite family **PROVED**

- equality with the arc value at $m \geq 3$ is open: an arc upper bound does not constrain the vertex-feasible family, which contains it



augmented bipartite digraph

ASYMPTOTICALLY EXTREMAL

COMPUTED Enumeration confirms $m = 2$ through $n = 7$, and the two separations agree in the completed $m = 3$ runs through $n = 6$.

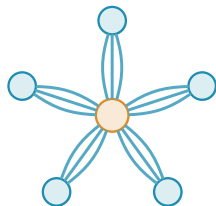
Multigraph · edge

$$L_m(n)$$

UPPER $(m-1)(n-1)$ **PROVED** a cut count, any shape

LOWER $(m-1)(n-1)$ **PROVED**

- exact for every n and m
- multiplicity removes the simple graph's distance-one restriction



tree, every edge repeated
 $m-1$ times **EXTREMAL**

COMPUTED Tabu search returns the multiplicity-2 star at $n=6$, $m=3$: $10 = L_3(6)$.

$$K_m(n)$$

UPPER $K_2(n) = n - 1$, $K_3(n) = 2(n - 1)$ **PROVED**

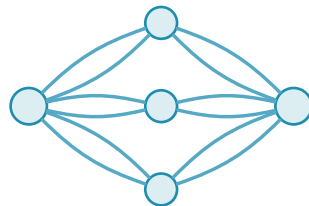
$$K_m(n) \leq (2 + o(1)) m^2 n$$
 PROPOSED

LOWER $K_3(n) \geq 2(n - 1)$ **PROVED**

$$K_m(n) \geq (3 - 2\sqrt{2} + o(1)) m^2 n$$
 PROPOSED

- the two coefficients bracket the growth as $m \rightarrow \infty$ with $n/m \rightarrow \infty$, and no common limit is established
- parallel copies are separated routes, so this is not the simple problem: $K_5(4) = 14$ against $L_5(4) = 12$

COMPUTED $K_5(4) = 14$ and $K_5(5) = 19$ by exhaustion. A nine-vertex sweep over all 191826 two-connected graphs with at least fourteen edges gives $g_6(9) = 54$.



thickened complete bipartite
block **NEAR-EXTREMAL**

Directed multigraph · arc

$$L_m^{\text{dir}}(n)$$

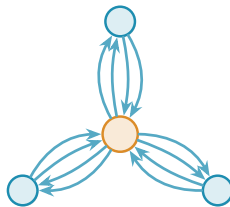
UPPER $(m-1)M(n)$ **PROVED** a sparse skeleton per unit of $m-1$

LOWER $(m-1)M(n)$ **PROVED** two branches, both attained

- exact for every n and m , and the fractional optimum is $M(n)$ too, so one m -free optimisation bounds every threshold
- the classification of the extremisers for $n \geq \max(8, 2m)$ stays

PROPOSED

COMPUTED Tabu returns the double star at $m=3$: 12 at $n=4$ and 16 at $n=5$.



thickened bidirected tree

EXTREMAL linear branch, $n \leq 7$

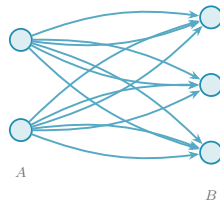
Directed multigraph · vertex

$$K_m^{\text{dir}}(n)$$

UPPER $(m-1)\frac{n^2}{4} + O_m(n)$ **PROVED**

LOWER $(m-1)\lfloor n^2/4 \rfloor$, thickened bipartite **PROVED**

- the arc extremiser is not even a finite optimum here:
 $K_6^{\text{dir}}(3) = 24$ against an established arc maximum of 20
- the exact correction and the best block shapes are open



thickened one-way bipartite
digraph

ASYMPTOTICALLY EXTREMAL

COMPUTED Direct enumeration gives $K_6^{\text{dir}}(3) = 24$.

Simple hypergraph · edge

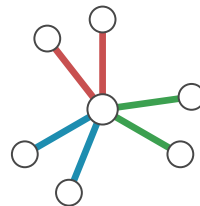
$$\ell_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** induction on a hyperedge cut

LOWER the same value **PROVED** when $m-1 \leq \binom{n-2}{r-2}$

- one hyperedge spans r vertices but serves only one route
- outside that range attainment is not guaranteed: at $n = r = m = 3$ the bound reads 2 and only one hyperedge exists

COMPUTED Exhaustive enumeration gives 11 at $(n, m, r) = (6, 6, 3)$, below the bound 12.



star hypertree ($m = 2$)

EXTREMAL

Simple hypergraph · vertex

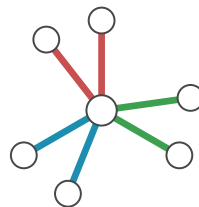
$$k_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ at $m = 2$, $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ at $m = 3$ **PROVED**
 $m \geq 4$: **OPEN**

LOWER star at $m = 2$, degree-capped hub at $m = 3$ **PROVED**
 $m = 3$ for $2 < r < n$

- $m = 2$ is feasible exactly when the incidence graph is a forest
- $m = 3$ comes from the incidence-rank inequality
- an edge bound does not transfer: $\kappa \leq \lambda$ points the wrong way, vertex feasibility being the weaker restriction

COMPUTED Enumeration at $m = 3$, $r = 3$ gives 1, 3, 4 for $n = 3, 4, 5$: the bound $\lfloor 2(n-1)/(r-1) \rfloor$ from $n = 4$ on, one short at $n = r = 3$.



star hypertree ($m = 2$)
EXTREMAL

Multihypergraph · edge

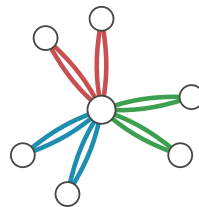
$$L_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** unchanged by repetition

LOWER $\frac{(m-1)(n-1)}{r-1}$ **PROVED** when $(r-1) \mid (n-1)$

- the constructions change even though the bound does not
- at $n = r = m = 3$, two copies of the only triple reach a value the simple model cannot

COMPUTED A twelve-copy construction at $(n, m, r) = (6, 6, 3)$ exceeds the simple edge optimum of 11.



star hypertree, edges repeated
 $m - 1$ times **EXTREMAL**

Multihypergraph · vertex

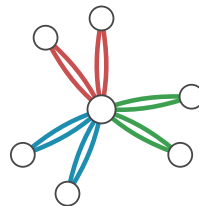
$$K_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ at $m = 2$, $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ at $m = 3$ **PROVED**
larger m : **OPEN**

LOWER the repeated star adds attainment cases at $m = 3$
PROVED

- repetition does not help at $m = 2$: two copies of one hyperedge are already two separated routes

COMPUTED The twelve-copy edge witness at $(6, 6, 3)$ is a vertex lower bound, and does not prove the vertex maximum is twelve. A sweep at $m = 4$ attains $\lfloor 3(n-1)/(r-1) \rfloor$ and excludes one more in eleven multihypergraph cases, all $r \geq 3$.



repeated star hypertree

EXTREMAL $m \leq 3$, and
 $(r-1) \mid (n-1)$ at $m = 3$

Directed simple hypergraph · arc

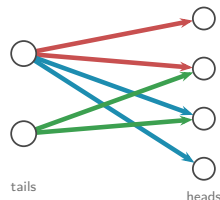
$$\ell_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED** shadow digraph

LOWER the same constant **PROVED** two-layer, $r \geq 3$

- the two meet, so the leading term is settled and the finite exact values are **OPEN**
- the matching argument covers distinct two-step routes sharing one hyperedge

COMPUTED At $n = m = 6$, $r = 3$, five distinct outgoing hyperedges per tail give 30, with a direct degree-cut feasibility check.



two-layer construction

ASYMPTOTICALLY EXTREMAL

Directed simple hypergraph · vertex

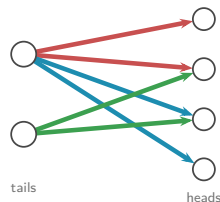
$$k_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED**

LOWER the same constant **PROVED** two-layer, $r \geq 3$, fixed m and r
as $n \rightarrow \infty$

- the shadow argument re-runs under the stricter separation, so the leading term is the same
- whether the finite edge and vertex maxima agree is **OPEN**

COMPUTED Random greedy growth with shuffled restarts measures what the implemented search finds, and establishes no optimum.



two-layer construction

ASYMPTOTICALLY EXTREMAL

Directed multihypergraph · arc

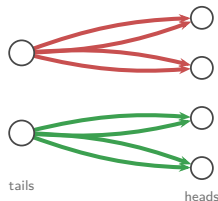
$$L_m^{\text{dir}}(r)(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED** forward, backward and
general alike

LOWER the same constant **PROVED** repeats lift the head-set limit

- the finite values can differ from the simple model, and are **OPEN**
- at $r = 2$ these are directed multigraphs, where the simple coefficient is $1/4$ instead, which is why $r \geq 3$ is kept

COMPUTED The plotted experiments use rank three and the forward convention. General orientations are checked separately in the appendix.



two-layer, hyperedges repeated

ASYMPTOTICALLY EXTREMAL

Directed multihypergraph · vertex

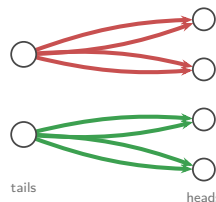
$$K_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED**

LOWER the same constant **PROVED**

- the same leading term under both separations
- finite exact values **OPEN**, and whether forward and general agree once $n > r$ is open too

COMPUTED The rank-three forward-convention runs are not an exhaustive comparison of the three orientation models.



two-layer, hyperedges repeated

ASYMPTOTICALLY EXTREMAL